Department of Mathematics

National Institute of Technology Kurukshetra

MAIC 207: Optimization Techniques (Mid Sem-1) Total Marks:15

Use of Calculators is allowed Time: 50 minutes

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1. (2 mark) A company produces two types of leather belts A and B. Belt A is of superior quality and belt B is of inferior quality. The respective profits are 10 and 5 per belt. The supply of raw material is sufficient for making at most 850 belts per day. For belt A, a special type of buckle is required and only 500 buckles are available per day. Belt B requires twice as much time as belt A, and the total time available per day is sufficient to produce 500 belts if all of them were of type A. Formulate a LPP model for this problem.

2. (5 mark) Use Simplex method to solve the following problem:

$$\begin{aligned} \text{Maximize } \ z &= 5x_1 - 4x_2 + 3x_3 \\ \text{subject to } 2x_1 + x_2 - 6x_3 &= 20 \\ 6x_1 + 5x_2 + 10x_3 &\leq 76 \\ 8x_1 - 3x_2 + 6x_3 &\leq 50 \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

3. (2 mark) Find the local maxima, local minima, and saddle points (if any) of the function

$$f(x, y) = \frac{1}{x} + xy + \frac{1}{y}, \quad (x \neq 0,\ y \neq 0).$$

4. (4 mark) Find the dimensions of a cylindrical tin (with top and bottom) made up of sheet metal to maximize its volume such that the total surface area is equal to 24π.

5. (2 mark) Verify whether the following optimization problem is convex or not:

$$\begin{aligned} \text{Maximize } \ z &= -4x_1 + {x_1}^2 - 2x_1x_2 + 2{x_2}^2 \\ \text{subject to } 2x_1 + x_2 &\leq 6 \\ x_1, x_2 &\geq 0. \end{aligned}$$